Medical Diagnosis and Report Generation for Endoscopic Procedures Using Vision-Language Models

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May 13, 2026

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2025

CERTIFICATE

Certified that this is a bonafide record of the project work titled

MEDICAL DIAGNOSIS AND REPORT GENERATION FOR ENDOSCOPIC PROCEDURES USING VISION-LANGUAGE MODELS

done by

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of eighth semester B. Tech in partial fulfillment of the requirements for the award of the degree of Bachelor of Technology in Computer Science and Engineering of the National Institute of Technology Calicut

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DECLARATION

We hereby declare that the project titled, **Medical Diagnosis and Report Generation for Endoscopic Procedures Using Vision-Language Models**, is our own work and that, to the best of our knowledge and belief, it contains no material previously published or written by another person nor material which has been accepted for the award of any other degree or diploma of the university or any other institute of higher learning, except where due acknowledgement and reference has been made in the text.

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Abstract

Background and Objective: Automated generation of endoscopic examination reports is a clinically significant yet technically demanding problem. Producing such reports demands deep familiarity with domain-specific terminology, pathological presentation, and the structured reasoning conventions of clinical practice, a burden that consumes considerable time even for experienced endoscopists. While multimodal large language models (MLLMs) have demonstrated impressive capabilities in joint visual and language processing, their utility in specialised medical domains is constrained by inadequate command of clinical vocabulary, difficulty recognising subtle pathological features, and limited adherence to medical reporting logic. Standard fine-tuning approaches partially address these gaps, yet commonly fail to provide the depth of domain-specific adaptation needed for high-quality endoscopic report generation. Methods: Building on the two-phase prompt-based LoRA (P-LoRA) framework introduced by Zhang et al. [42], we develop MVLMERG (Multimodal Vision-Language Model for Endoscopic Report Generation). The first stage enriches the base model with broad medical domain knowledge sourced from publicly available datasets, while the second stage specialises the model for clinical endoscopic reporting through fine-tuning on physician-annotated data. Results: On our curated endoscopic dataset, MVLMERG achieves state-of-the-art results, surpassing all evaluated baseline systems across every automated evaluation metric.

ACKNOWLEDGEMENT

This work is supported by the Department of Computer Science and Engineering, National Institute of Technology Calicut.

We sincerely thank our project guides, **Dr. Jayaraj PB** (Associate Professor) and **Srinivasa TM** (Assistant Professor), for their invaluable guidance, continuous support, and encouragement throughout the course of this project.

We are deeply grateful to the following investigators from **Government Medical College, Kozhikode**, for providing the clinical dataset and their expert medical insights:

1. **Dr. Sandesh K**, MD DM, Professor, Head of Department
2. **Dr. Annavaram Jeshwanth Reddy**, MD

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Introduction

Project Information

Title: Medical Diagnosis and Report Generation for Endoscopic Procedures Using Vision-Language Models.
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Keywords: Multimodality; Large language models; Fine-tuning; Vision-language models; Endoscopic report generation.

1.1 Introduction

Endoscopic examinations occupy a central role in the early identification, diagnosis, and clinical management of gastrointestinal conditions [3,4], and a growing body of work has continually advanced the state of the art for associated computational tasks [1,2]. Every procedure produces substantial volumes of visual data that clinicians must interpret meticulously before composing accurate written reports, a process that is simultaneously cognitively demanding and time-intensive. Multimodal large language models (MLLMs) offer a compelling path forward: by pre-training on broad-scale corpora, they acquire transferable general knowledge that can subsequently be directed toward narrow specialised domains, making them well-suited to bridging the gap between raw endoscopic imagery and structured diagnostic text [5].

Constructing unified multimodal systems for clinical practice has led researchers to explore architectures capable of integrated cross-modal reasoning [6,7]. Large biomedical repositories such as PubMed [8] furnish extensive general medical knowledge, yet their coverage of pathology-specific or organ-level annotations remains limited. The scarcity of carefully labelled endoscopic image data further constrains the development of robust diagnostic systems. Compounding these data-availability challenges, building large multimodal architectures from the ground up demands prohibitive computational investment; consequently, parameter-efficient adaptation of pre-existing models has emerged as a far more tractable strategy [9].

Although LoRA demonstrates broad transferability across diverse tasks, its adaptation performance on highly specialised downstream problems can fall short of clinical requirements [11]. Combining prompt-based conditioning mechanisms with LoRA inside the LLaVA-4 framework therefore constitutes a pressing open challenge for domain-specific deployment.

Following Zhang et al. [42], whose staged fine-tuning methodology directly addresses these limitations, this project constructs MVLMERG (Multimodal Vision-Language Model for Endoscopic Report Generation) as a targeted clinical solution. The training corpus is built from physician-annotated endoscopic records subsequently standardised with GPT-5. Adopting the two-phase P-LoRA framework of [12,42], MVLMERG first undergoes broad medical domain enrichment that equips the model with the clinical vocabulary and conceptual grounding required to reason accurately about endoscopic findings. A second fine-tuning stage then steers the model toward producing outputs that conform to the structural and stylistic conventions of professional endoscopic documentation.

The principal contributions of this work are as follows:

1. A specialised vision-language system, MVLMERG, purpose-built for clinical endoscopic report generation, which outperforms Qwen2-VL-7B-Instruct, BLIP-2, and InstructBLIP across every automated evaluation metric on our institutional endoscopic dataset.

2. A two-phase prompt-guided LoRA (P-LoRA) fine-tuning strategy that explicitly separates the acquisition of medical domain knowledge from the learning of professional report-generation conventions, enabling effective adaptation of LLaVA-4 to the endoscopic reporting task with substantially reduced computational overhead.

The remainder of this report proceeds as follows. Chapter 2 surveys the relevant literature on medical documentation burden, multimodal large models, and parameter-efficient fine-tuning. Chapter 3 formally defines the problem addressed by this work. Chapter 4 presents the MVLMERG architecture and methodology in full, covering the two-stage fine-tuning pipeline, the P-LoRA mechanism, dataset construction and preprocessing, image augmentation procedures, and chain-of-thought reasoning integration. Chapter 5 reports experimental findings and quantitative comparisons against baseline systems. Chapter 6 draws conclusions and outlines directions for future research.

Chapter 2

Literature Survey

2.1 Evolution of Medical Documentation Burden

Over recent decades, administrative documentation tasks have come to occupy a growing share of clinical working hours, with primary care physicians bearing a disproportionate part of this burden [40]. The 2019 National Electronic Health Records Survey provides concrete measurements: physicians spend an average of 1.77 hours per day on documentation activities conducted outside of scheduled office time, and this figure rises to 1.84 hours among those who use EHR systems, against only 1.10 hours for non-EHR users [41].

The effects reach further than time loss alone. Surveys indicate that 75% of healthcare providers consider current documentation demands to be actively harmful to the quality of care they can deliver [40]. At a more granular level, 58.1% of physicians reject the view that their documentation workload is appropriately sized or that it preserves sufficient time for patient interaction, and 84.7% acknowledge that record-keeping driven purely by billing obligations inflates their total documentation hours [40,41]. Collectively, these figures make a compelling case for intelligent automation tools that can absorb documentation overhead while maintaining the accuracy and professional standards required in clinical records.

2.2 MLLMs

Large-scale pretraining has given rise to multimodal large language models (MLLMs) that jointly process visual and textual information, dramatically broadening the scope of AI perception and generation. A central design challenge in this area is achieving deep, rather than superficial, coupling between vision and language components. Wang et al. [13] tackled this problem through CogVLM, which inserts trainable visual expert modules directly into the attention and FFN layers of a frozen language backbone, enabling parameter-level fusion of visual and linguistic representations without degrading natural language processing behaviour. Approaching the same challenge from a different angle, Bai et al. [14] introduced the Qwen-VL model family to remedy the difficulty that standard language models face when interpreting raw visual signals; a three-stage curriculum training procedure applied to multilingual data allowed Qwen-VL to set new performance marks on image captioning, visual question answering, and visual grounding evaluations. Liu et al. [15] pursued yet another strategy, exploiting GPT-4's text-only reasoning capacity to synthesise multimodal instruction-following data, yielding LLaVA, a system that couples a vision encoder with a large language model backbone and achieves comparable results to multimodal GPT-4 on images unseen during training. Despite these advances, clinical data constitute only a small fraction of the corpora used to train such systems, leaving general-purpose models poorly equipped to characterise lesion morphology accurately or to produce outputs that conform to the structured conventions of professional diagnostic reporting.

2.3 Medical MLLMs

Parallel to the growth of general-purpose vision-language research, the healthcare domain has steadily accumulated heterogeneous multimodal data (spanning clinical images, operative recordings, histological slides, structured laboratory results, and longitudinal health records) that now underpin a new generation of medically specialised MLLMs. Concerning the quality and breadth of medical image–text supervision, Chen et al. [16] constructed the PubMedVision dataset by harvesting image–text pairs from PubMed and subjecting them to an MLLM-assisted denoising step; training HuatuoGPT-Vision on this curated resource produced marked gains across a range of medical multimodal benchmarks. In surgical settings, Seenivasan et al. [17] observed that the unidirectional autoregressive nature of GPT-style decoders prevents direct ingestion of surgical image inputs, and responded with SurgicalGPT, an end-to-end trainable architecture that resolves this constraint and achieves accuracy improvements of 3%–5% across three surgical evaluation sets. For colonoscopy in particular, Ji et al. [18] identified a gap in both task-appropriate benchmarks and existing model capability, addressing it by releasing the ColonINST benchmark alongside the ColonGPT model, which surpassed prior approaches on colonoscopy classification, referring expression, and localisation while demonstrating stronger cross-task generalisation.

2.4 Fine-tuning methods

The prohibitive cost of pretraining large multimodal models from scratch, combined with the performance degradation that results from deploying unadapted general models on narrow clinical tasks, has established task-specific fine-tuning as the dominant adaptation strategy for medical MLLMs [20,21]. Li et al. [7] provide a prominent illustration with LLaVA-Med, which approaches biomedical image understanding via a two-stage pipeline: an initial alignment phase on the PMC-15M corpus anchors the model in biomedical vocabulary, after which GPT-4-generated instruction data drives full end-to-end instruction tuning. Concerned with the parameter overhead that medical MLLM adaptation typically incurs, He et al. [22] proposed PeFoMed, a framework whose two-stage fine-tuning design achieves competitive results on medical imaging benchmarks without proportionally expanding the set of trainable parameters. Chen et al. [23] focused on the twin obstacles of data privacy restrictions and scarce expert labels that frequently hinder medical visual question answering, introducing a generative pretraining and fine-tuning pipeline designed to operate effectively under these constraints. Lu et al. [24] demonstrated that casting visual features as soft visual prompts aligned with a language model's text embedding space, combined with a two-stage fine-tuning schedule, yields meaningful improvements in both the accuracy and generation efficiency of radiology report synthesis. In pathology, Wu et al. [25] assembled PathEnhanceDS, a 45,000-case instruction-tuning corpus, and showed that it raises model competence on pathological image classification, descriptive captioning, and question answering tasks. Notwithstanding these contributions, prevailing fine-tuning approaches share three persistent shortcomings: (1) clinical domain knowledge is insufficiently embedded during the fine-tuning process, leaving models with shallow comprehension of specialised medical semantics; (2) existing systems lack flexible, task-aware prompting mechanisms that can steer model attention toward the most diagnostically salient visual features; and (3) the practice of applying LoRA uniformly across every transformer layer is computationally wasteful, since it ignores the functionally distinct roles that initial, intermediate, and terminal layers each play in multimodal representation learning.

Chapter 3

Problem Definition

Automated endoscopic report generation sits at the intersection of computer vision and clinical natural language generation, and presents challenges that general-purpose multimodal models are ill-equipped to handle. Physicians who interpret endoscopic examinations must integrate visual findings with specialised clinical knowledge and produce structured reports conforming to established medical conventions, a process that is both cognitively demanding and time-consuming. General-purpose MLLMs, despite their broad cross-modal capabilities, frequently misidentify pathological features, misuse domain terminology, and produce outputs that lack the structured reasoning required in endoscopic practice. Following Zhang et al. [42], who identified these limitations and proposed a staged fine-tuning strategy to address them, this project adopts a similar problem framing: effective endoscopic report generation requires not merely task-specific fine-tuning but a deliberate injection of medical domain knowledge followed by alignment to professional reporting conventions. The objective of this project is therefore to design and implement MVLMERG, a system capable of bridging the knowledge gap of general vision-language models, incorporating clinically relevant domain understanding, and generating accurate, structured endoscopic reports that meet professional medical standards.

Chapter 4

Methodology

4.1 Method

MVLMERG is realised by applying the P-LoRA adaptation scheme [42] to LLaVA-4, the chosen backbone, within a curriculum that proceeds across two sequential fine-tuning phases. The sections below describe each architectural and training component in detail.

4.1.1 Overview

Figure 1 illustrates the end-to-end training architecture of MVLMERG. The system is trained in two sequential phases. During the first phase, a curated corpus of publicly available endoscopy-related text (including diagnostic reports, prescriptions, clinical dialogues, and analogous material) supplies the primary source of domain knowledge; P-LoRA weight updates are applied to the language model while all vision encoder and connector parameters are held constant. The second phase is carried out with the involvement of specialist gastroenterologists at Government Medical College Kozhikode, who provided an in-house multimodal corpus of gastric endoscopy images alongside expert-authored clinical descriptions. In this phase the same freeze policy is maintained; only the language model weights receive gradient updates, directing the system toward the vocabulary and report structure characteristic of formal clinical documentation. This staged regime, following the design proposed in [42], progressively integrates visual and linguistic domain knowledge while keeping computational overhead low. The open-source MLLM LLaVA-4 [39] serves as the backbone of our framework. Visual features are derived from gastric endoscopic images using the pretrained CLIP-ViT-L/14 encoder [15,39]. An input RGB image $I \in \mathbb{R}^{H \times W \times 3}$, with spatial dimensions $H \times W$, is processed by the CLIP-ViT-L/14 visual encoder, yielding a sequence of visual embeddings $\mathbf{V} = [\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_{N_v}]$ of length N_v. These embeddings are projected into the language model's representation space and prepended to the tokenised text sequence $\mathbf{T} = [\mathbf{t}_1, \mathbf{t}_2, \ldots, \mathbf{t}_{N_t}]$ of length N_t. The joint sequence is processed by the Vicuna-7B-v1.5 decoder, which autoregressively produces the output response $\mathbf{R} = [\mathbf{r}_1, \mathbf{r}_2, \ldots, \mathbf{r}_{N_r}]$ of length N_r.

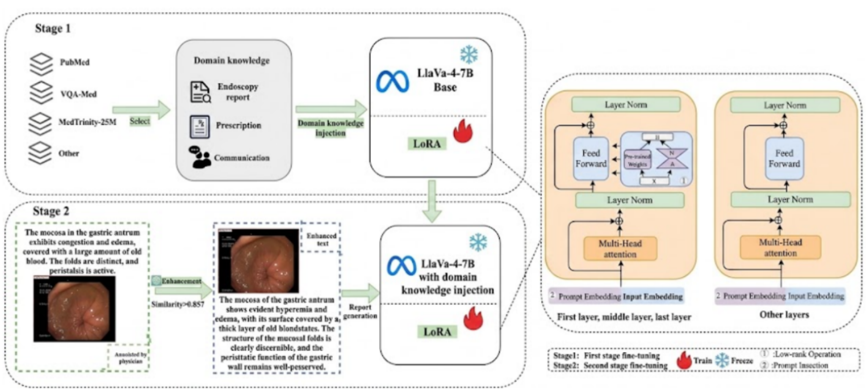


Figure 4.1: Overall architecture of the proposed two-phase fine-tuning pipeline. The left portion depicts Phase 1 (public-dataset domain injection) and Phase 2 (private clinical fine-tuning); the right portion provides a detailed view of the P-LoRA adaptation module. Green-highlighted components indicate the elements introduced in this work.

4.1.2 Two-Phase Fine-Tuning Strategy

Standard MLLMs lack sufficient exposure to endoscopy-specific knowledge, leaving significant gaps in their capacity to interpret lesion morphology and gastroenterological terminology. Following Zhang et al. [42], we address this deficit through a two-phase fine-tuning strategy that separates general medical domain enrichment from targeted clinical report alignment. The model first consolidates broad gastroenterological knowledge before being directed toward the professional register and structural conventions of endoscopic documentation, yielding outputs that are both medically accurate and clinically consistent. Phase 1 draws on three complementary publicly available datasets that collectively supply the gastroenterological breadth required for downstream specialisation.

PubMed [8] indexes more than 35 million biomedical citations and abstracts from life science journals worldwide. Our training material is sourced from the PubMed Central (PMC) Open Access subset, which supplies full-text articles together with embedded figure–caption pairs. A curated gastroenterology vocabulary is applied to retain only GI-relevant image–text records spanning the esophagus, stomach, duodenum, and colon. This filtered subset establishes semantically rich grounding in anatomical expressions, lesion terminology, and clinical descriptors that form the foundation for multimodal pretraining.

MedTrinity-25M [26] is a large-scale collection of approximately 25 million image–text pairs organised as (image, region, enriched description) triplets across diverse clinical modalities, including radiology (X-ray, CT, MRI), histopathology slides, and endoscopy. Each triplet associates a localised image region with a detailed, grounded textual description, supporting fine-grained visual–language alignment. The clinical breadth and scale of this dataset make it particularly suited to exposing the model to diverse pathological appearances and their corresponding diagnostic vocabulary.

VQA-Med [27], derived from the ImageCLEF 2019 medical VQA challenge, provides over 4,200 radiology images each paired with targeted clinical question–answer items spanning four question types: modality recognition, plane identification, organ identification, and abnormality detection. Exposure to this dataset develops the model's capacity to associate image-level visual evidence with structured clinical reasoning, strengthening its response accuracy on domain-specific queries and complementing the descriptive knowledge furnished by PubMed and MedTrinity-25M.

Together, these three datasets supply foundational knowledge of gastric anatomy, lesion localisation, tissue-level structural changes, and standard diagnostic terminology. Throughout this phase, vision encoder and connector weights are frozen, and parameter updates are restricted to the language model alone. In Phase 2, physician-annotated records from Government Medical College, Calicut are used to align the model's language generation with real-world clinical reporting practice. The data undergo the targeted processing pipeline described in Section 4.1.4 before fine-tuning. This phase prioritises accurate command of professional terminology, discipline-consistent sentence structure, and the specific expression patterns characteristic of formal endoscopic reports, producing outputs that faithfully reproduce the documentation standards expected in clinical settings.

4.1.3 P-LoRA Method

While LoRA achieves broad generalisation across tasks, its adaptation performance on narrow domain-specific problems can be insufficient when applied in isolation [11]. As proposed by Zhang et al. [42], our P-LoRA method addresses this limitation by coupling learnable soft prompt tokens with layer-selective LoRA weight updates within a unified parameter-efficient framework. Soft prompts are continuous, task-optimised embedding vectors prepended to the input sequence; unlike hard prompts, which encode manually crafted discrete tokens, soft prompts reside directly in the embedding space and are refined end-to-end through gradient descent with no manual engineering required. This mechanism guides the model toward domain-appropriate representations at minimal parameter cost. Selective LoRA weight increments then provide complementary, targeted adjustments that sharpen feature extraction across both visual and textual streams. Neither component alone is sufficient: soft prompts without weight updates may fail to reshape deep internal representations, whereas unrestricted LoRA incurs higher parameter overhead. Following Zhang et al. [42], LoRA is applied selectively at three structural positions in the network: the initial layer, the intermediate layers, and the terminal layer. This reflects the empirically observed principle that initial layers encode general low-level features, intermediate layers mediate cross-modal interactions, and terminal layers specialise for task-specific output. Learnable soft prompt tokens [28] are prepended to the input at every attention layer and initialised from the same distribution as the pretrained token embeddings. Let $\mathbf{e} = (e_1, e_2, \ldots, e_m)$ denote the trainable prompt tokens and $\mathbf{h} = (h_1, h_2, \ldots, h_n)$ the original input embeddings, where each $h_i \in \mathbb{R}^d$ and d is the hidden dimension. The augmented sequence $\mathbf{z} = (e_1, \ldots, e_m, h_1, \ldots, h_n)$ is fed into the attention layer, where queries, keys, and values are computed jointly over all positions. The prompt tokens steer attention toward domain-relevant features while leaving the pretrained backbone weights unchanged; they are optimised exclusively through gradient descent on the task objective. Low-rank weight updates via LoRA are subsequently applied to the feed-forward layers. Restricting LoRA to the first, middle, and final Transformer layers serves distinct functional purposes: the initial layer is adapted to realign the model's sensitivity to novel domain-specific input patterns; the intermediate layers are updated to recalibrate attention over gastroenterological features; and the terminal layer is tuned to condition output generation on task-specific requirements. Within each selected feed-forward block, the first linear projection expands the representation from d_{model} to d_{ff}, and a low-rank increment is introduced at this weight matrix. Rather than updating the full weight matrix $W \in \mathbb{R}^{m \times n}$

as in conventional fine-tuning, the pretrained weights are frozen and only an additive term ΔW is learned, giving the adjusted matrix shown in Formula (1).

$$W' = W + \Delta W \tag{4.1}$$

The increment is factorised as $\Delta W = BA$, with $B \in \mathbb{R}^{m \times r}$, $A \in \mathbb{R}^{r \times n}$, and rank $r \ll \min(m, n)$, so only the compact low-rank factors are trained. This factorisation captures the most task-discriminative directions in weight space while drastically reducing the number of trainable parameters. A ReLU non-linearity applied to the intermediate activations introduces further representational depth, after which the second projection maps the expanded d_{ff} representation back to d_{model}, restoring dimensional consistency with the rest of the network. The feedforward sub-layer thereby combines the non-linearly enriched representations with the contextual signals accumulated by the attention layer, strengthening the model's overall expressiveness. Through this architecture, the attention and feedforward sub-layers play complementary roles: prompt-augmented attention layers heighten the model's responsiveness to domain-specific cues, while LoRA-updated feedforward layers amplify representational richness through non-linear low-rank transformations. The combined effect yields strong task adaptation with substantially reduced training cost. Given image I and prompt P, the model generates a diagnostic report $Y = (y_1, y_2, \ldots, y_T)$ via the autoregressive decoding objective in Formula (2).

$$Y = \arg\max \prod_{t=1}^{T} P(y_t \mid y_{<t}, I, P; \theta) \tag{4.2}$$

where θ denotes all model parameters and $y_{<t}$ represents the sequence of tokens generated prior to position t.

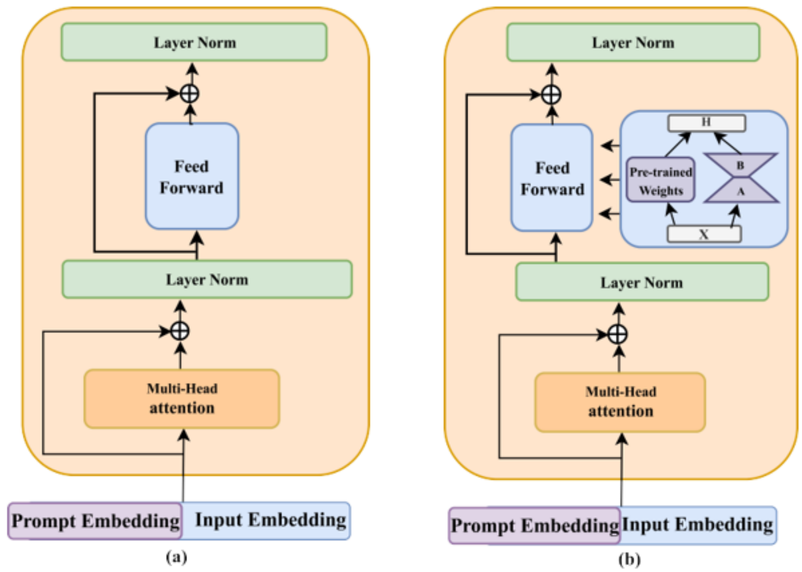


Figure 4.2: Schematic of the P-LoRA adaptation strategy. Panel (a) depicts the soft-prompt insertion applied to intermediate transformer layers; panel (b) shows the selective LoRA injection applied at the initial, intermediate, and terminal layers of the language model backbone.

4.1.4 Data Optimisation and Filtering with GPT-5

The private training corpus for Stage 2 consists of clinical gastroscopy reports obtained from Government Medical College, Calicut. Each record documents a patient's upper gastrointestinal examination, accompanied by endoscopic images spanning the following anatomical regions: Esophagus; GE Junction; Stomach (Fundus, Body, Antrum); Pylorus; Duodenum. While the images supply the direct visual evidence required for model training, the physician-authored narratives provide the clinical grounding necessary for report generation. In practice, however, a proportion of records contain incomplete entries, non-standard phrasing, or minor transcription errors. Given the well-documented text-processing capabilities of GPT-5 [29,30], we adopt it as an auxiliary annotation tool to refine and enrich these raw sentences. Imposing a uniform template structure across all training descriptions serves two purposes: it eliminates noise introduced by heterogeneous writing styles, and it ensures the model's attention during fine-tuning remains on clinically meaningful content rather than incidental formatting variation. For the templatisation step, GPT-5 receives a concise structured prompt instructing it to reformat each description into the template: *"Region: [anatomical site]. Findings: [colour, texture, lesion presence]."*. Quality control is governed by a three-step pipeline: raw text is first submitted to GPT-5 for template conversion; each generated output is then independently reviewed by experienced gastroenterologists, who assess medical accuracy, correct terminology usage, and logical coherence; outputs failing these criteria are revised or removed. Under this workflow, GPT-5 acts strictly as a preprocessing assistant while domain-expert physicians retain final authority over clinical validity. An invariant constraint governing every optimisation step is that the semantics of the revised sentence must be fully preserved; no clinical information may be added, omitted, or distorted in the reformulation. To enforce this requirement, a cosine similarity measure over sentence embeddings is employed. Sentence-BERT [31] encodes both the original and the reformulated sentence into fixed-dimensional embeddings; denoting these embeddings as $\mathbf{a}$ and $\mathbf{b}$ respectively, the semantic consistency score σ is computed as shown in Formula (3).

$$\sigma = \frac{\mathbf{a} \cdot \mathbf{b}}{\|\mathbf{a}\| \|\mathbf{b}\|} \tag{4.3}$$

where σ denotes the similarity score. A threshold of 0.85 is applied: any reformulated sentence scoring below this value is rejected and regenerated until the criterion is satisfied, ensuring that data enrichment does not compromise the clinical fidelity of the training corpus.

4.1.5 Data Augmentation

Clinical endoscopic records sourced from Government Medical College, Calicut yielded an initial collection of approximately 6,500 physician-annotated reports. Although each record meets high clinical quality standards, this volume alone is inadequate for robust fine-tuning of a large multimodal model. To expand the dataset while preserving clinical validity, two complementary image augmentation strategies are applied: rotation augmentation and stochastic translation augmentation.

Rotation Augmentation

Each original endoscopic image is rotated by 0°, 90°, 180°, and 270°, producing four variants per image. This is clinically safe because tissue textures and lesion appearances are invariant to rotation in endoscopic imaging. The augmentation produces a structured 4× expansion of the base dataset:

- Base reports: 6,500
- After rotation: $6{,}500 \times 4 = 26{,}000$ reports

Each report consists of 960 tokens (784 visual tokens + 176 text tokens), so the rotated corpus amounts to $26{,}000 \times 960 = 24{,}960{,}000$ tokens.

Stochastic Translation Augmentation

To simulate the natural camera jitter that occurs during real endoscopic examinations, we apply stochastic translation to a subset of the rotated images using a Bernoulli selection process with probability $p = 0.2$. For each selected image, the translation vector (τ_x, τ_y) is drawn from a truncated Gaussian distribution:

$$\tau_x \sim \mathcal{N}(0,\, (0.05W)^2), \quad \text{clamped to } [-0.15W,\, +0.15W]$$

$$\tau_y \sim \mathcal{N}(0,\, (0.05H)^2), \quad \text{clamped to } [-0.15H,\, +0.15H]$$

where W and H are the image width and height respectively. This guarantees that at least 72.25% of the original image content is preserved, ensuring that diagnostic features remain intact. The translation augmentation adds approximately $26{,}000 \times 0.2 = 5{,}200$ additional reports, contributing $5{,}200 \times 960 = 4{,}992{,}000$ tokens.

Combined Dataset Size

The combined token count after both augmentations is:

$$24{,}960{,}000 + 4{,}992{,}000 \approx 29{,}952{,}000 \text{ tokens} \approx 29.95\text{M tokens}$$

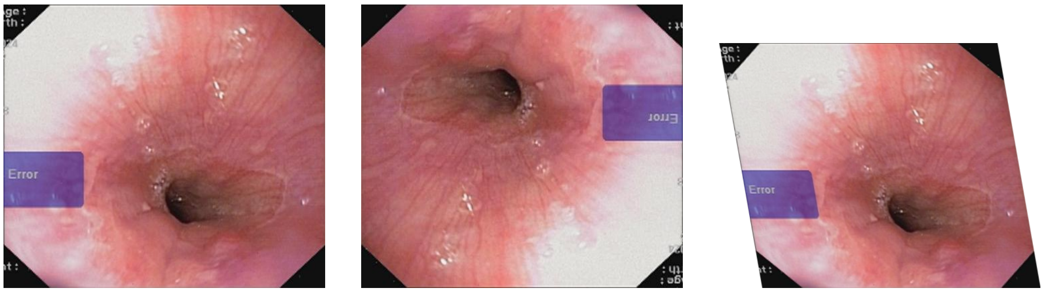
The final second-stage training set thus comprises approximately 31,200 image–report pairs, which substantially expands the original 6,500 clinical reports while preserving clinical validity. A 5% held-out validation split (per the training configuration) is applied to this augmented dataset, yielding approximately 29,640 training samples and 1,560 validation samples.

Figure 4.3 illustrates representative examples of the two augmentation strategies applied to a sample endoscopic image.

4.1.6 Chain-of-Thought Integration

The Interpretability Problem and Motivation

When a model generates diagnostic labels or report text directly from an endoscopic image without exposing its intermediate reasoning, the output is opaque: clinicians have no way to verify or interrogate the inferential steps that led to the conclusion. This “black-box” behaviour is a fundamental obstacle to clinical adoption. To mitigate it, we augment the second-stage training data with chain-of-thought (CoT) reasoning sequences that make the model’s decision pathway explicit.



(a) Original (b) Rotated (90°) (c) Stochastic Translation

Figure 4.3: Representative augmentation examples drawn from the GMC Calicut corpus. (a) Original endoscopic frame. (b) Frame after 90° rotation; rotation invariance of gastric lesion morphology makes this transformation clinically safe. (c) Frame after stochastic translation sampled from a truncated Gaussian ($\sigma = 5\%W$, Bernoulli trial $p = 0.2$), preserving at least 72.25% of the original image content.

CoT Implementation

Every training question–answer pair is reformulated to include an explicit step-by-step reasoning chain. The chain imposes a fixed structure on the model's response: *anatomical identification* → *visual observation* → *diagnosis*. Through fine-tuning on these augmented samples, the model learns to predict the complete reasoning chain rather than only the final conclusion, making its inferential process transparent and auditable.

For example, contrast the two modes of response:

Before CoT (baseline):

> *Input:* [Image] "What is the diagnosis?"
> *Output:* "Grade II Esophageal Varices."

After CoT (our model):

> *Input:* [Image] "Analyze the image step-by-step and provide a diagnosis."

Output:
Observation 1: I observe the lower third of the oesophagus.
Observation 2: There are swollen, tortuous submucosal veins present.
Observation 3: The veins occupy less than one third of the lumen, with no red colour signs.
Conclusion: Therefore, the diagnosis is Grade II Esophageal Varices.

Impact on Clinical Utility

Incorporating CoT supervision converts the system from an opaque classifier into an interpretable diagnostic assistant. Three concrete clinical benefits follow: (1) each reasoning step can be verified independently by the reviewing clinician; (2) erroneous intermediate observations become visible and can be corrected before they propagate to the final conclusion; and (3) the structured output format mirrors the conventions of formal clinical documentation, reinforcing medically appropriate language generation. These properties are especially important in real-world deployment settings, where clinical trust and professional accountability are non-negotiable requirements.

Chapter 5

Results

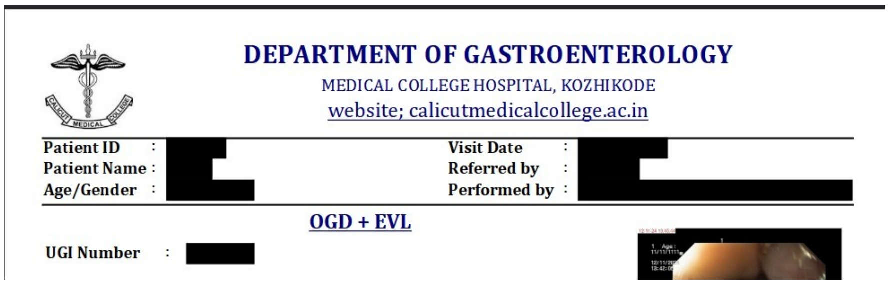
5.1 Experiments and discussion

To validate the effectiveness of our proposed method, the model was first pre-trained on publicly available medical datasets and subsequently fine-tuned on our institutional private dataset; performance is evaluated across automated metrics and expert physician assessment.

5.1.1 Datasets

The training corpus is assembled through a two-phase procedure, with the first phase drawing gastrointestinal descriptive content from publicly available medical datasets, as summarised in Table 5.1. A curated gastroenterology terminology list drives automated query-based retrieval, restricting the collected samples to material that is directly relevant to the target clinical domain and excluding records whose subject matter falls outside gastrointestinal medicine. Each candidate report subsequently undergoes an independent quality review by practicing gastroenterologists, who discard samples that are clinically ambiguous, incomplete, or otherwise unsuitable for supervised training, thereby elevating the overall reliability and professional standard of the retained corpus. The underlying sources, namely PubMed [6], MedTrinity-25M [12], and VQA-Med [6], span a broad range of medical topics and together offer detailed organ-level descriptions alongside extensive disease-specific terminology. The resulting collection furnishes the model with grounded knowledge of gastrointestinal anatomy, characteristic lesion presentations, and domain-appropriate semantic patterns, establishing the representational foundation required for the more specialised second-phase training. The second phase relies on upper gastrointestinal endoscopy reports sourced from Government Medical College, Calicut, which are selected and preprocessed according to the procedure outlined in Section 4.1.4 to yield a tightly focused training set. All records originate from actual clinical encounters and must pass through a purpose-built automated de-identification pipeline before entering the training corpus; this step is mandated by India's Digital Personal Data Protection (DPDP) Act and aligns with HIPAA-equivalent privacy requirements. The pipeline systematically and irreversibly eliminates every Protected Health Information (PHI) field from the text body and its associated metadata: patient identifiers such as name and patient ID are removed, as are age and gender fields, encounter dates, and procedure reference numbers (UGI-OGD report numbers). Because the removal is cryptographically one-way, re-identification from the sanitised records is not feasible. Figure 5.1 illustrates this pipeline as applied to a representative patient report header.

Within the assembled dataset, approximately 40% of cases are classified as normal, with the remaining 60% representing pathological findings; this deliberate imbalance towards abnormal cases ensures the model is exposed to sufficient pathological diversity for discriminative learning. The pathological category covers a broad spectrum of gastrointestinal diagnoses, among them esophageal varices, mucosal polyps, gastrointestinal bleeding, peptic ulcers, and early-stage and advanced gastric malignancy. Coverage of the gastrointestinal tract is systematic, with images spanning the Esophagus, GE



DEPARTMENT OF GASTROENTEROLOGY

MEDICAL COLLEGE HOSPITAL, KOZHIKODE

website; calicutmedicalcollege.ac.in

Patient ID : [redacted]

Patient Name : [redacted]

Age/Gender : [redacted]

Visit Date : [redacted]

Referred by : [redacted]

Performed by : [redacted]

OGD + EVL

UGI Number : [redacted]

Figure 5.1: Automated PHI removal pipeline for GMC Calicut clinical records. Prior to any use in model training, the system irreversibly suppresses all identifying fields, including Patient ID, full name, age and gender, procedure dates, and UGI-OGD reference numbers, to satisfy the requirements of India's Digital Personal Data Protection (DPDP) Act.

Junction, Stomach (Fundus, Body, Antrum), Pylorus, and Duodenum, guaranteeing both anatomical comprehensiveness and clinical representativeness. All frames are selected intraoperatively by attending endoscopists from diagnostically relevant views during routine examinations. Acquisition hardware varies across endoscopic platforms available at Government Medical College, Calicut, yet a uniform native resolution of 1920×1080 is maintained throughout. For model input, the 1920×1080 captures are downsampled to 336×336 in the preprocessing stage prior to ingestion by the CLIP-ViT-L/14 vision encoder. Training and validation partitions follow a 5% validation split, while a fully independent held-out test set is reserved exclusively for final performance evaluation. Conformity with the standardised acquisition protocol means that all retained images satisfy clinical diagnostic standards with respect to image clarity, anatomical coverage, frame validity, and lesion visibility. To guard against terminological drift or stylistic inconsistency introduced during data optimisation, multiple senior physicians conduct rigorous post-processing reviews of the finalised reports, correcting any deviations in medical accuracy or professional register. Quality verification confirms that all retained records satisfy the predefined similarity criterion, with 78.6% of entries attaining a similarity score above 0.9, a result that reflects a high overall standard of report consistency and reliability. The quality-controlled collection comprises approximately 6,500 physician-annotated endoscopy reports originating from Government Medical College, Calicut. These records are characterised by consistent professional register, reflecting the vocabulary choices, syntactic patterns, and diagnostic conventions that experienced endoscopists employ in formal reporting practice; these attributes are directly transferable to the model's generative output in the target clinical domain. Although the second-stage corpus is substantially smaller than the first-stage public data, its value lies in concentrated clinical specificity: every record is devoted to precise visual description of endoscopic findings and structured documentation of lesion characteristics.

5.1.2 Implementation details

All fine-tuning experiments are built on the LLaVA-4 base model, with P-LoRA applied identically in both training stages. The adapter configuration is held constant across stages: rank = 64, $\alpha = 32$, and dropout = 0.05. Each stage processes data in micro-batches of size 2; four gradient-accumulation steps bring the effective global batch size to 8. A peak learning rate of 2e-4 is adopted together with a warm-up ratio of 0.1, and the token budget per sequence is capped at 4096. Four training epochs are completed under a cosine decay schedule, employing the AdamW optimiser with 8-bit quantisation (adamw_bnb_8bit) and bfloat16 (BF16) mixed precision throughout. Flash Attention 2 is activated to lower the memory cost of self-attention over long contexts. The random seed is fixed at 42 to ensure reproducibility. Greedy decoding (temperature = 0, top_p and top_k both disabled) is used at inference time; the full hyperparameter listing appears in Table 2.

Vision Encoder CLIP-ViT-L/14. Trained on 400M image-text pairs, this encoder delivers reliable medical visual representations; the L-scale variant strikes a favourable balance between representational capacity and computational throughput, while the 14×14 patch decomposition simultaneously encodes broad anatomical context and fine-grained lesion detail.

Image resolution 336×336 pixels. This resolution is the native input size for LLaVA-family models and preserves clinically relevant detail at a memory footprint compatible with multi-sample batching.

Patch size 14×14 (24×24 grid).

Total vision tokens 577 per image (576 patch tokens plus one CLS token). The per-patch tokenisation grants the language decoder spatially grounded access to both coarse-grained anatomical layout and localised region-level cues.

Training Precision bfloat16 (BF16). BF16 cuts memory consumption roughly in half relative to FP32 while maintaining numerical stability throughout training.

Attention Flash Attention 2. An exact-attention algorithm that achieves 5–10× VRAM savings for extended token sequences by restructuring memory access patterns.

All experiments are executed on a Linux machine running PyTorch, equipped with dual Intel Xeon 6505P processors (12 cores per socket, 2 sockets, 48 threads in total), 251 GiB of system RAM, and a single NVIDIA H200 NVL GPU (143 GB HBM3e, CUDA 13.0). The software environment consists of Python 3.12.3 and PyTorch 2.11.0+cu130. By applying P-LoRA, only 0.343% of the total model parameters are marked as trainable, leaving the backbone frozen and substantially reducing storage and compute overhead.

Table 5.2: Training hyperparameters for Phase 1 and Phase 2 fine-tuning runs.

Hyperparameter	Phase 1	Phase 2
Learning rate	2e-4	2e-4
Optimizer	AdamW-8bit	AdamW-8bit
Warm-up ratio	0.1	0.1
Num epochs	4	4
Random seed	42	42
Batch size	2	2
Gradient accumulation	4	4
Max sequence length	4096	4096
LoRA rank	64	64
α	32	32
Dropout	0.05	0.05

5.1.3 Evaluation metrics

Following the evaluation protocols established in prior works [32–34], we adopt CIDEr, ROUGE, and METEOR as primary automated metrics to quantify the correspondence between generated reports and ground-truth reference descriptions. CIDEr (Consensus-based Image Description Evaluation) [35] was formulated specifically for image captioning benchmarks. Its distinguishing characteristic is a TF-IDF weighting scheme applied at the n-gram level: n-grams that recur frequently across the corpus are down-weighted as uninformative, whereas rarer, domain-specific phrases receive elevated scores, producing assessments that correlate better with human judgements of caption quality. ROUGE (Recall-Oriented Understudy for Gisting Evaluation) [36] provides an automatic quality estimate for text generation systems and has become standard in summarisation, translation, and captioning evaluation pipelines. The metric quantifies how much content from the reference text is recovered by the generated output, making recall the central quantity of interest. METEOR (Metric for Evaluation of Translation with Explicit Ordering) [37] was developed to remedy known deficiencies in BLEU [38]-class metrics, which inadequately account for morphological variation and paraphrase relations between a candidate and its reference. METEOR combines exact token matching with stem- and synonym-level alignment, yielding sensitivity to both lexical form and semantic equivalence across natural language generation tasks. Because automated metrics capture surface-level textual similarity rather than clinical correctness or diagnostic utility, they alone are insufficient for evaluating medical reports; accordingly, structured expert physician evaluation is incorporated as a complementary assessment.

5.1.4 Experimental results

Comparison of different models

To establish the relative merit of the proposed approach, a set of widely used multimodal large language models, namely Qwen2-VL-7B-Instruct, BLIP-2 (opt-2.7b), and InstructBLIP, is selected as representative baselines, and a comprehensive quantitative comparison is presented in Table 3. Evaluation is conducted on a randomly sampled subset of 100 held-out validation cases. Uniform evaluation conditions are maintained by providing each baseline with the identical input prompt used by our model: "You are a medical endoscopy report assistant. Respond with only the clinical finding using correct medical terminology. Be concise.". Following Zhang et al. [42], the proposed model diverges from the aforementioned baselines by being systematically adapted to the specialised task of endoscopic report generation through targeted fine-tuning of a powerful foundation backbone. The two-phase curriculum (general gastrointestinal knowledge acquisition in the first phase, then professional clinical report generation in the second) allows the model to build domain understanding incrementally rather than attempting to acquire both capacities simultaneously. Complementing this staged schedule, the task-specific architectural design and the domain-tailored fine-tuning protocol together confer markedly improved adaptability to the endoscopic imaging domain. Equally important is the construction of the high-quality, physician-annotated institutional corpus, which provides the precise professional style and clinical grounding that generic datasets cannot supply.

Table 5.1: Training data composition across both fine-tuning phases. For each source, the table identifies the dataset origin, content modality, record volume, and a concise characterisation of its role in the training curriculum.

Training Phase	Source Dataset	Content Type	Sample Volume	Notes
Phase 1 (General Pre-training)	PubMed; MedTrinity-25M; VQA-Med	GI-domain knowledge: image–caption and text pairs	60k	Open-source medical corpora covering broad gastrointestinal anatomy, lesion semantics, and diagnostic vocabulary; used to establish domain grounding prior to clinical fine-tuning.
Phase 2 (Clinical Fine-tuning)	Physician-annotated endoscopy records, Government Medical College Kozhikode	Institution-specific endoscopic report data	6.5k (base); ∼31k (augmented)	De-identified, professionally validated reports capturing the register and diagnostic conventions of trained gastroenterologists. Expanded four-fold via systematic rotation and stochastic translation augmentation.

Table 5.3: Quantitative comparison of MVLMERG against baseline vision-language models. All scores are computed on a randomly sampled subset of 100 held-out validation cases across five automated metrics: BLEU-4, METEOR, ROUGE-L, CIDEr, and BERTScore.

Method	BLEU-4 ↑	METEOR ↑	ROUGE-L ↑	CIDEr ↑	BERTScore ↑
Qwen2-VL-7B-Instruct	0.0009	0.0564	0.0504	0.0078	0.8317
BLIP-2	0.0002	0.0347	0.0147	0.0079	0.7957
InstructBLIP	0.0004	0.0583	0.0320	0.0088	0.8196
Our Model	**0.0006**	**0.1187**	**0.2632**	**0.0531**	**0.8760**

Chapter 6

Conclusion and Future work

6.1 Limitations and future work

For example, outstanding open problems include automated pinpointing of lesion boundaries, reliably grading and classifying gastric malignancies, and accurately interpreting atypical or overlapping lesion presentations. Despite the model's strong overall accuracy in report generation, the outputs are not error-free. Clinically consequential mistakes, such as failure to report early malignancy or active haemorrhage, can directly alter management decisions and therefore represent the most critical failure mode. By contrast, errors that amount to minor phrasing differences or stylistic variation carry little clinical weight. Physician assessment confirmed that the large majority of mistakes belong to the latter, lower-risk category; nonetheless, even rare high-consequence errors underscore why every model-generated report must be reviewed by a qualified clinician before it informs patient care. Taken together, these observations reinforce the principle that AI-assisted tools should be embedded within, rather than substituted for, established clinical review processes. Training on data drawn from a single institution introduces a risk that performance degrades when the model is applied in other hospital settings. Across institutions, differences in the make and model of endoscopes, the lighting and colour-rendering characteristics of each system, and the varying vocabulary and diagnostic thresholds adopted by individual clinicians can all shift the statistical properties of both images and reports. When the training corpus predominantly reflects the annotation habits of a small cohort of clinicians, the resulting model may inadvertently encode their particular choice of wording, their relative emphasis on certain findings, or their tolerance for diagnostic uncertainty. Regional variation in disease prevalence and case complexity compounds this risk, as differences in patient demographics and clinical practice patterns across sites can further shift the statistical properties of both images and reports, degrading generalisation to unseen settings. When visual evidence is ambiguous or internally inconsistent, the model is designed to produce conservative, finding-based descriptions anchored to directly observable features rather than committing to a definitive diagnosis, thereby limiting the potential for clinical over-interpretation. Systematic uncertainty quantification—enabling the model to communicate its own confidence alongside each generated report—and automated routing of low-confidence cases to senior clinician review constitute high-priority directions for future development. The present study was conducted at a single institution, and the most immediate next step is a structured multi-centre evaluation. We plan to partner with at least two hospitals in different regions of India that operate distinct endoscope platforms and serve demographically varied patient cohorts. Performance will be measured on each site's held-out split both before and after brief site-specific P-LoRA fine-tuning, quantifying the adaptation cost required to sustain clinical-grade accuracy across differing imaging environments and reporting conventions. Beyond site-level generalisation, the system currently processes individual endoscopic frames in isolation. In clinical practice, gastroenterologists review several sequential images before committing to a report, drawing on patterns of lesion persistence and morphological change across frames. Extending the input context to ordered image sequences would allow the model to generate integrated patient-level summaries that more closely mirror this workflow. A longer-horizon direction is full clip-level temporal modelling, enabling the detection of transient events (such as active bleeding, peristaltic irregularities, or instrument-tissue interactions) that remain invisible in any single still frame. Enlarging the institutional dataset is a standing priority. We are in active discussion with the Department of Gastroenterology at Government Medical College Kozhikode to prospectively annotate a further 2,000–3,000 endoscopy cases, concentrating on under-represented categories including early Barrett's oesophagus, sub-centimetre polyps, and post-endoscopic submucosal dissection healing patterns. Subject to clearance from the Institutional Ethics Committee (IEC) and compliance with India's Digital Personal Data Protection Act, it is hoped that a substantially larger annotated corpus from Government Medical College Kozhikode will become available for subsequent training cycles. Following IEC approval, a de-identified subset will be released to support reproducibility and further research in automated endoscopic reporting. The trained model weights, P-LoRA adapter configurations, and training scripts will similarly be made openly available upon publication. On the deployment front, the model will be assessed under INT8 and INT4 weight quantisation to verify that compression does not erode sensitivity to clinically consequential findings. An HL7 FHIR-compatible inference API and a lightweight integration module are planned to connect the system to the electronic medical records infrastructure at Government Medical College Kozhikode without disrupting established clinical workflows. Full deployment will proceed only after formal approval from the relevant Institutional Ethics Committee and satisfaction of any Software as a Medical Device (SaMD) requirements applicable under Indian regulatory frameworks, including guidelines issued by the Central Drugs Standard Control Organisation (CDSCO).

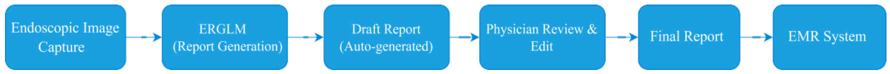


Figure 6.1: Intended deployment workflow for MVLMERG in a clinical setting: an endoscopic image is ingested by the system, which produces a structured draft report; a reviewing clinician edits and approves the draft before it is committed to the electronic medical record (EMR).

6.2 Conclusion

Building on the P-LoRA framework introduced by Zhang et al. [42], this study constructs an endoscopic image interpretation system that combines prompt-conditioned low-rank adaptation with a two-stage fine-tuning curriculum, designed to unlock the latent diagnostic capacity of pre-trained multimodal foundation models. Following both fine-tuning phases, the resulting model produces clinically detailed endoscopic descriptions and demonstrates notably strong sensitivity to morphological image characteristics. Across all quantitative benchmarks, the proposed system achieves substantially higher scores than the evaluated comparative models, including Qwen2-VL-7B-Instruct, BLIP-2, and InstructBLIP.

CRediT authorship contribution statement

Praval Pattam: Software, Writing – original draft, Methodology, Conceptualization. Theenesh Potluri: Software, Writing – review & editing, Validation, Formal analysis. Pranav Sai Sarvepalli: Software, Data curation, Investigation.

Ethical Considerations

- **Informed Consent:** Written informed consent will be obtained from all prospective patients.

- **Patient Privacy:** Complete de-identification of all patient data will be performed according to the strict protocol detailed in Section 3.8.7. Data handling will comply with the Digital Personal Data Protection Act (DPDP Act), Government of India, and institutional data security policies.

- **Clinical Oversight:** All AI-generated reports will be reviewed and approved by qualified endoscopists.

- **Safety and Risk Mitigation Protocol:** The AI system will not influence real-time clinical decision-making during the validation phase. All AI-generated reports will undergo mandatory review and approval by a qualified endoscopist prior to finalisation. AI outputs will be clearly labelled as decision-support only. All outputs will be logged for audit, and discrepant cases will undergo review by an oversight committee.

- **Data Ownership and IP:** A formal Memorandum of Understanding (MoU) between Government Medical College Kozhikode and NIT Calicut will define data ownership, intellectual property rights, publication rights, and technology transfer terms prior to project initiation. Data sharing between Government Medical College Kozhikode and NIT Calicut will occur under the project-specific collaboration agreement executed under the umbrella MOU dated 27 April 2022. All raw clinical data including endoscopic images and reports remain the property of Government Medical College Kozhikode. De-identified datasets will be shared with collaborating investigators solely for the purposes of this study. Furthermore, the final trained model will be made available to the institution for local control and will not be exclusively hosted externally.

Declaration of competing interest

The authors confirm that no financial interests, personal relationships, or affiliations exist that could be perceived as influencing the outcomes or conclusions presented in this work.

Data availability

The data underlying this study cannot be shared due to institutional restrictions and patient privacy obligations.

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